Assignment_LAB→02

Ques(1):- What are the differerd protocols you observe at following layers of the protocls stack?

(A) Application layer
(B) Transport Layer
(C) Network layer.

Ans→ (A) Application layer protocls :-

- HTTP
- TLS/HTTPS,
- DNS
- mDNS
- QUIC

(B) Transport layer Uses:-

- TCP
- UDP

(C) Networt layer protocol used :-

- IPV4
- IPV6
- ICMP
- ICMPV6
- ARP

Ques(2):→ What is the total amount of deta being received for following two cases?

(A) When you access http://neverssl.com
Ans→ 2273 Byte

(B) When you access https://www.cornell.edu
Ans→ 3,795,640 Bytes.

Que(3):- Answer when you access the side http://neverssl.com/

(A) How many HTTP GET requests do you observed list down the GET request.

Ans→ GET / HTTP/1.1
Host: neverssl.com
Connection: keep-alive
Upgrade-Insecure-requests: 1

(B) What version of HTTP the server is running

Ans→ HTTP/1.1

(C) For which of the ~~following~~ HTTP GET requests, find out:

(i) Total TCP segments received in the response.

Ans→ 5 data carring segments
(Packets # 17056 - # 17060)

(ii) Total data received in the HTTP response message.

Ans→ 2,273 byte = 373-byte Header + 1,900-byte gzip body.

(8)(D) Identify TCP three way handshake used to establish the connection with the HTTP://neverssl.com. specify SYN, SYN-ACK, ACK packets and write the sequence no. for each of the three accesses.

Ans→

Connection	SYN (seq)	SYN-ACK (seq/ack)	ACK (seq/ack)
#1 (Client port 60299)	seq = 1394760637	seq = 2170695920/ ack = 1394760638	seq = 1394760638 ack = 2170695921
#2 (Client Port 56925)	seq = 3234514881	seq = 1741309461 ack = 3234514882	seq = 3234514882 ack = 1741309462
#3 (Client port 64283)	seq = 3649373126	seq = 1295756439 ack = 3649373127	seq = 3649373127 ack = 1295756440